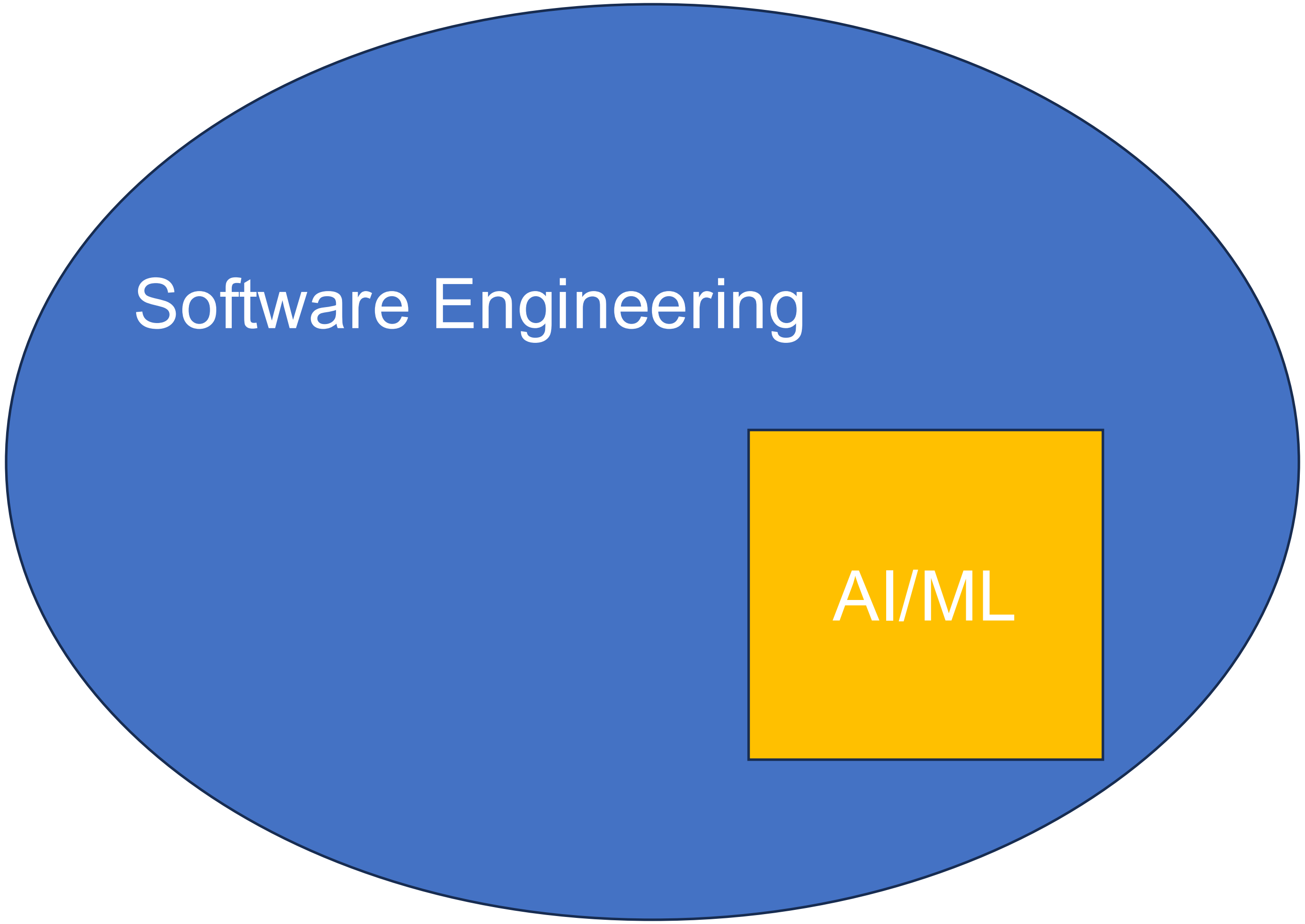
Lecture 2

AI-CR-503

Digital Technology Program

Madan Bhandari University of Science and Technology (MBUST)

UNIT 2 Software engineering perspective for AI/ML



Software engineering (SE)

- Software Engineering (SE) is defined as the process of designing, developing, testing, and maintaining software.

- A variety of techniques, tools, and methodologies are implemented by software engineers for the analysis, design, testing, and maintenance of software.

What is AI based systems?

- A system consisting of various software and potentially other components, out of which at least one is an AI component.

- It can be studied under three dimensions such as scope, application domain, and technologies of AI.

a) Scope: It refers to how AI is implemented in the system. AI can either be one component in a system, dominating the entire system, implement one or more particular algorithms, or providing a pipeline or infrastructure.

b) Application domain: Sometimes application domain of AI is not clearly defined, but often it is more specific for example autonomous driving which also include autonomous driving system.

c) Technologies of AI: It refers to the core of AI such as machine learning in general or deep learning method.

Difference between software and AI/ML system

S.N.	Software system	AI/ML system
1.	Output is fixed. (Deterministic)	Output depends on probability. (Probabilistic)
2.	Test for correctness.	Statistical test for performance.
3.	All requirement is defined.	Requirements evolve with data.
4.	Code inspection is necessary.	Model testing and analysis.

Traditional software

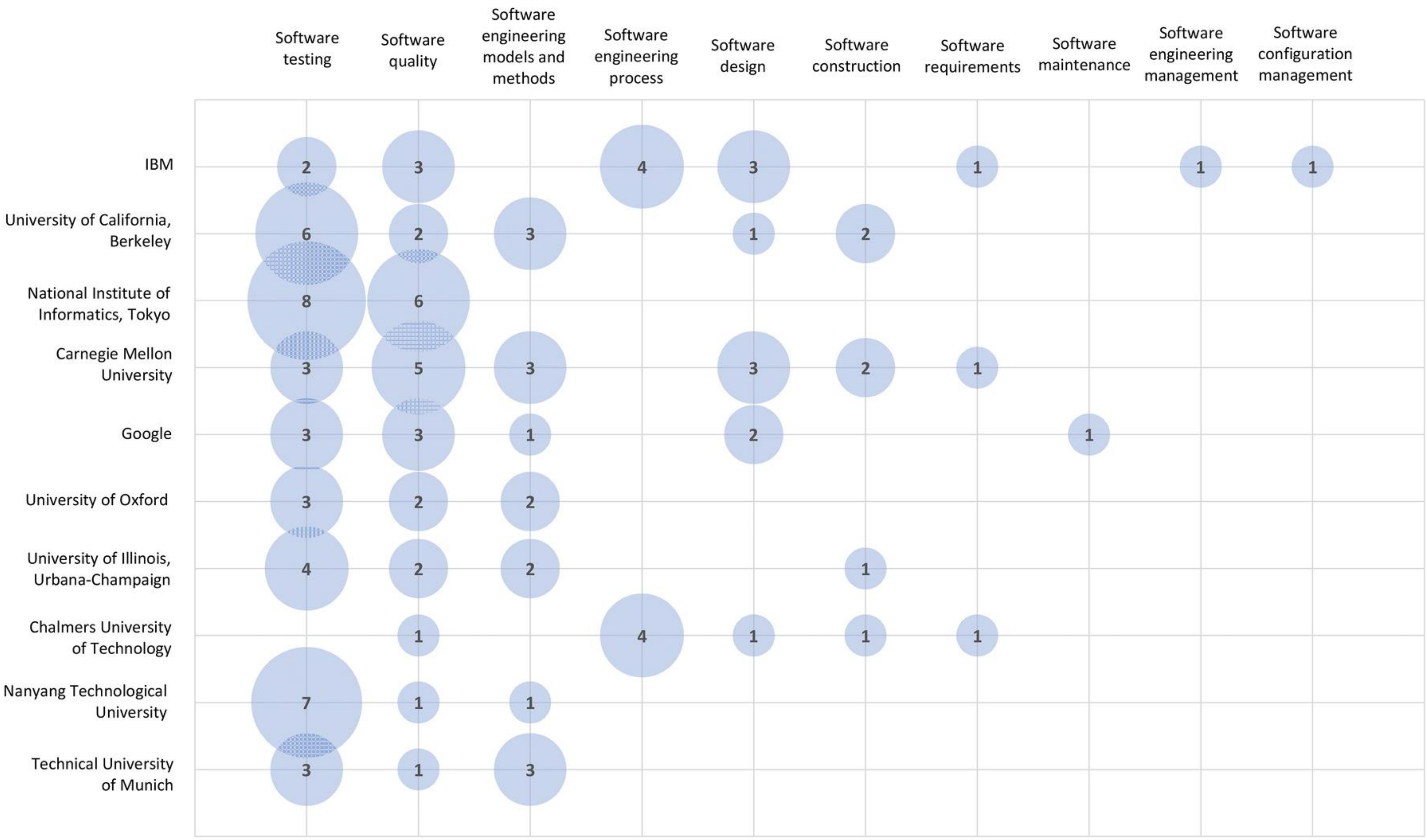
Machine learning

Software engineering for AI (SE4AI)

- AI and SE are related, as they both have their roots in computer science.
- It refers to addressing various SE tasks for engineering AI systems, i.e., designing, developing, and maintaining AI-enabled software systems.
- Researchers are trying to identify different aspects of engineering AI systems compared to traditional software and develop new techniques and tools to cope with these differences.

SE approaches for AI-based systems

- There are 11 areas where SE contributed to AI-based systems.
    1. Software requirements.
    2. Software design.
    3. Software construction.
    4. Software maintenance.
    5. Software engineering models and methods.
    6. Software engineering process.
    7. Software quality.
    8. Software testing.
    9. Software engineering management.
    10. Software configuration management.
    11. Software engineering professional practice.



Leading institutions using SE approaches for AI-based systems.

Challenges of SE approaches for AI-based systems

1. Challenges are highly specific to the AI/ML domain. Many of the SE approaches are not applied completely and there is artifacts in architectural and design patterns.

2. Challenges are mainly of a technical nature: It refers to identifying deficiency in the knowledge areas such as computing, mathematics, and engineering.

3. Data-related issues are the most recurrent type of challenge: Examples are preparing high-quality training datasets, identifying important features in big datasets, and management of datasets.

Thank you.